Xingjian (Scott) Zhang

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ML engineer for reliable, efficient LLM systems: KV-cache inference, retrieval-augmented generation and rigorous evaluation.

Seeking a Summer 2027 internship (available May–Aug 2027).

Education

Carnegie Mellon University Pittsburgh, PA
M.S. in Artificial Intelligence Engineering – Electrical & Computer Engineering *Aug 2026 – Dec 2027 (expected)*
Coursework: Machine Learning for Engineers; Deep Learning for Computer Vision; Systems & Toolchains for AI Engineers

New York University New York, NY
B.A. in Computer Science, Minors in Mathematics & Economics; GPA: 3.7/4.0 *May 2026*

Research Experience

Independent Researcher (sole author) – efficient and reliable LLM inference 2026 – Present
- **SAFE-KV** (Findings of AACL-IJCNLP 2026) [paper] [code]: showed that non-prefix KV-cache reuse in RAG keeps only 47.2%/37.8% of full-prefill answers (HotpotQA/2Wiki, Qwen2.5-7B); designed causal suffix repair that restores 99.8%/98.4% and cuts changed-wrong answers from 241 to 1; released with 135 unit tests.
- Further sole-author work is under double-blind review, spanning KV-cache compression, long-context RAG, LLM evaluation and test-time compute; details will be posted once review concludes.

Efficient Sequence Modeling – with Prof. Eunsol Choi & Yilun Kuang (NYU) May 2025 – Jan 2026
- Designed attention architectures between quadratic attention and constant-memory linear RNNs, and built a FLOPs-matched training/evaluation pipeline on Slurm (PyTorch, Triton).

LLM Reasoning Benchmark – with Prof. Haohan Wang (UIUC) Feb 2025 – Jun 2025
- Co-built AdvGameBench (arXiv:2506.12012), a strategic-game benchmark of how LLMs plan and revise under resource budgets; defined process metrics (over-correction risk, correction success, over-budget rate) tied to win rate.

Chart Question Answering – with Prof. Qingyun Wang (William & Mary) May 2025 – Jan 2026
- Worked on mixture-of-experts vision-language models for chart reasoning; built verified Program-of-Thought training data for SFT and DPO/GRPO post-training.

Transport Safety Modeling – with Dr. Faan Chen (Harvard) Jan 2024 – Sep 2025
- Built hybrid ML + multi-criteria decision models (tuned DBSCAN/K-means, GMM clustering) on a curated 2010–2023 road-safety panel (EU27+3, G20, East Asia Summit): 4 peer-reviewed journal papers (2 first-author; 60+ citations) and 4 open-source, single-command reproduction packages.

Publications

- X. Zhang. SAFE-KV: Causal Suffix Repair for Behavior-Preserving Non-Prefix KV Cache Reuse in RAG. *Findings of AACL-IJCNLP 2026*; arXiv:2610.15186.
- X. Yuan, X. Zhang, et al. Tracing LLM Reasoning Processes with Strategic Games. arXiv:2506.12012, 2025.
- Journals: X. Zhang et al., *Sci. Rep.* 15:36467 (2025); X. Zhang et al., *Results Eng.* 29:108543 (2026); Z. Guo, ..., X. Zhang, et al., *Knowl.-Based Syst.* 301:112286 (2024); J. Jin, ..., X. Zhang, et al., *Sci. Rep.* 15:35099 (2025).

Skills

ML & LLM: PyTorch, Hugging Face, vLLM, Triton; RAG, LoRA, SFT, DPO/GRPO
Programming & data: Python (pandas, NumPy, SciPy), C++, Java; SQL (PostgreSQL, Spark SQL), PySpark
Infrastructure: Linux, Docker, Git, Slurm/HPC; cloud GPUs (A100/H100), GCP, AWS
Evaluation & statistics: benchmark design, LLM-as-judge evaluation, bootstrap CIs, paired significance tests
Languages: Mandarin (native), English (fluent)